

Unseen Seminar Questions 4: Answers

Probability and Statistical Inference

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1. An analyst reports $\hat{\beta} = 0.42$ and $\widehat{se}(\hat{\beta}) = 0.15$. For the two-sided test of $H_0 : \beta = 0.10$, use a standard normal approximation, $\Phi(2.13) = 0.9834$ and $z_{0.975} = 1.96$. The analyst makes four claims:
- (a) the test statistic is approximately 2.13;
 - (b) the two-sided p -value is approximately 0.0332;
 - (c) the null is rejected at the 5% level;
 - (d) the 95% confidence interval is $[0.126, 0.714]$ and contains the null value.

Which claims can be true at the same time? Correct any error, and explain the general consistency check connecting a two-sided test with its matching confidence interval.

Solution. The statistic is

$$T = \frac{0.42 - 0.10}{0.15} = 2.133 \dots,$$

so (a) is correct. The two-sided p -value is approximately $2[1 - \Phi(2.13)] = 2(0.0166) = 0.0332$, making (b) correct. Since this is below 0.05 (and $2.13 > 1.96$), (c) is correct. The interval calculation is also correct:

$$0.42 \pm 1.96(0.15) = 0.42 \pm 0.294 = [0.126, 0.714].$$

However, $0.10 < 0.126$, so the interval does *not* contain the null value. Claim (d) must be corrected accordingly. For matching two-sided level- α tests and $(1 - \alpha)100\%$ confidence intervals, rejection occurs exactly when the null value lies outside the interval.

2. Rewrite each statement so that it has a correct frequentist interpretation.
- (a) “Since the observed p -value is 0.31, the planned test has 31% power.”
 - (b) “If we doubled the sample size, the p -value for the data already observed would automatically be cut in half.”
 - (c) “There is a 95% probability that the fixed parameter lies in this particular 95% confidence interval.”

Solution.

- (a) “The value 0.31 summarizes how unusual this sample is under the null. Power is the probability of rejection at a specified alternative, calculated before observing the data; it cannot be read from one observed p -value.”
- (b) “A larger sample in a new study may reduce its standard error and increase power, but it does not alter the data or p -value already observed, and there is no general rule that doubles n and halves p .”
- (c) “Across repeated samples generated by the same procedure, 95% of intervals constructed this way cover the fixed parameter.” Once this sample has produced its endpoints, the frequentist interval either covers the fixed parameter or it does not.

3. A pollution-control coefficient is estimated as $\hat{\gamma} = -0.28$ with standard error 0.10. Before seeing the data, the researcher specified $H_0 : \gamma = 0$ against $H_1 : \gamma < 0$. Using a standard normal reference distribution, $z_{0.95} = 1.645$, and $\Phi(-2.80) = 0.0026$:

- (a) carry out the test at the 5% level using both a critical value and a p -value;
- (b) **Non-examinable extension:** construct the matching 95% one-sided confidence set;
- (c) **Non-examinable extension:** explain why the infinite end of that confidence set is on the left rather than the right.

Solution.

- (a) The statistic is $T = (-0.28 - 0)/0.10 = -2.80$. A left-sided 5% test rejects when $T < -1.645$, so it rejects. Its p -value is $\Phi(-2.80) = 0.0026 < 0.05$, giving the same decision.
- (b) **Non-examinable extension.** The matching upper one-sided confidence set is

$$(-\infty, \hat{\gamma} + z_{0.95}\hat{se}] = (-\infty, -0.28 + 1.645(0.10)] = (-\infty, -0.1155].$$

It excludes the null value 0.

- (c) **Non-examinable extension.** Large negative values are compatible with the pre-specified direction $\gamma < 0$, so the set has no finite lower endpoint. The data instead supply an upper confidence bound. Null values above this bound, including zero here, are rejected in favor of a smaller coefficient.

4. Match each description to one of the three classical test methods: *Wald*, *likelihood ratio*, or *score/LM*.

- (a) Measures the distance from the unrestricted estimate to the null value.
- (b) Compares fit under the null restriction with fit under the unrestricted model.
- (c) Measures the slope of the objective function (the criterion being optimized) at the null value.

Then state the core testing logic shared by all three methods.

Solution.

- (a) Wald.
- (b) Likelihood ratio.
- (c) Score/LM.

For each method, the distribution of its chosen measure of evidence is derived under H_0 , and the null is rejected only when the observed evidence is sufficiently extreme. The statistics differ, but all three ask the same question: whether the null is compatible with the data.

5. **Choosing standard errors.** For each error pattern below, choose a suitable standard-error method from Lecture 4: conventional homoscedastic, heteroscedasticity-robust (White), HAC, or cluster-robust. Briefly explain your choice. Treat the patterns as conditional on the regressors. Assume the other conditions for the chosen method hold, with enough observations, time periods or independent groups for any large-sample approximation.

- (a) Errors have equal variances and are independent across observations.
- (b) Errors are independent across observations, but their variances differ.
- (c) Observations are ordered in time, and errors can be correlated across nearby periods.
- (d) People are sampled in groups. Errors may be correlated within each group, with different variances, but groups are independent of one another.

Solution.

- (a) Conventional homoscedastic standard errors: equal variances and zero cross-observation covariances give $\Omega = \sigma^2 I_n$.
- (b) White standard errors: they allow heteroscedasticity while observations remain independent.
- (c) HAC standard errors: they allow heteroscedasticity and serial correlation under the technical conditions assumed in the question. White standard errors alone omit serial covariances.
- (d) Cluster-robust standard errors, using the sampling groups as clusters. These allow unequal variances and correlation within groups. The approximation requires enough independent groups, not just many people in a few groups.